SMART INDIA HACKATHON 2026

PS – SIH26073

SkyGuard AI

Intelligent Real-Time Anomaly Detection for Automatic Weather Stations

AI/ML-based intelligent anomaly detection for AWS — for the India Meteorological Department (IMD), Ministry of Earth Sciences (MoES).

THEME:
Disaster Management

CATEGORY:
Software

TEAM NAME:
Darth Coders

Proposed Solution

Core question: how do we distinguish abnormal weather from abnormal sensors?

Triangulated Anomaly Detection Paradigm

Statistical Detector

Range checks, rate-of-change, deviations & persistence vs. historical behaviour.

Spatial Detector

Consistency with nearby stations via Inverse Distance Weighting (IDW).

LSTM Autoencoder

Learns normal multivariate temporal behaviour; flags high reconstruction error.

Addressing Core Operational

- **Multi-Detector Fusion Engine**

  Eliminates single point of failure by taking majority/confidence consensus, stopping false storm alerts from polluting forecaster feeds.

- **Frozen Calibration State**

  Guarantees reproducible, drift-free inference across AWS production stations without model degradation or silent inference shifts.

- **Historical & Synthetic Benchmarking**

  Rigorous verification with synthetic anomaly injection against known ground truth to systematically validate edge-case recall.

- **Nationwide AWS Scalability**

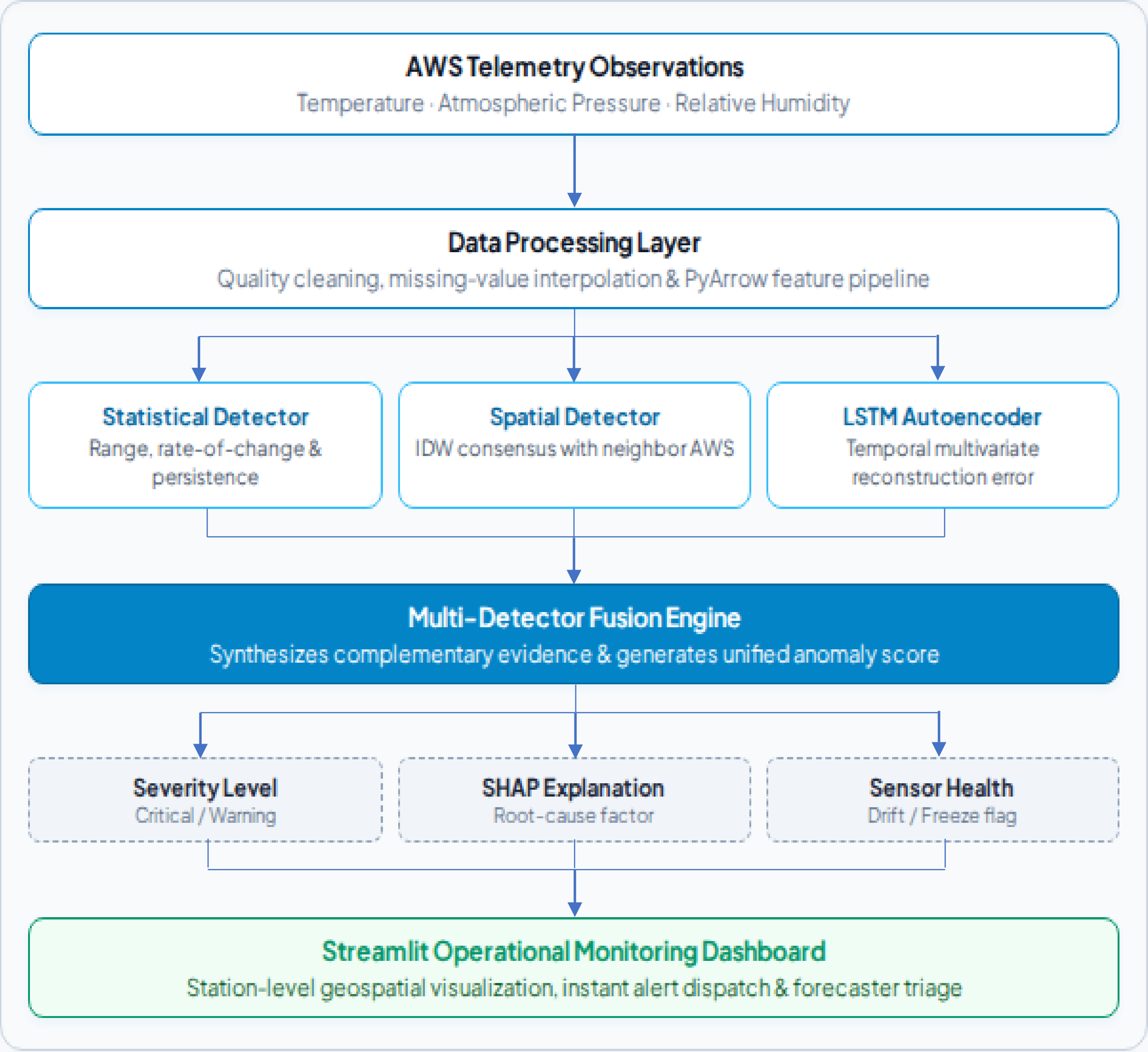
  Decoupled microservice architecture designed for seamless horizontal scaling across hundreds of IMD regional weather stations.

Technical Approach

Tech Stack

as implemented in the current build

1. **Python**
   Core language for the pipeline
2. **Pandas · NumPy · PyArrow**
   Data processing & feature pipeline
3. **Scikit-learn**
   Statistical detector implementation
4. **TensorFlow / Keras**
   LSTM Autoencoder detector
5. **SHAP · LIME**
   Explainable anomaly contributors
6. **Plotly · Matplotlib**
   Station visualization & charts
7. **Streamlit**
   Dashboard — deployed on Streamlit Community Cloud



Feasibility & Viability

Challenge	Risk	Mitigation Strategy
Distinguishing genuine weather from sensor/data anomalies	Misclassification propagates errors into forecasting & disaster-management systems	Fusion Engine combines Statistical, Spatial & LSTM Autoencoder evidence for a single anomaly decision
Observation-level detection is imperfect (F1 = 53.68%)	556 false positives / 678 false negatives in the current benchmark	Event-level recall reaches 89.06% (114/128 events) — most real anomaly events are still caught
No live AWS data-stream access yet during development	Cannot validate directly against operational feeds	Evaluated via controlled synthetic anomaly injection with known ground truth for systematic accuracy measurement
Scaling beyond the current pilot deployment	Live integration & national rollout add complexity	Architecture built for scalable deployment; live AWS stream integration is a defined next step

89.06%

event recall — 114 of 128 anomaly events detected

53.68%

F1 score (any-two-of-three fusion)

0.817%

false positive rate across all observations

1,271

alerts raised — 1.83% alert rate

Impact & Benefits

S SOCIAL
Faster, more trustworthy disaster alerts

Severity-scored, explainable anomaly alerts help IMD forecasters and disaster-management teams trust AWS data during critical weather events.

E ECONOMIC
Lower manual QC & maintenance cost

Automated detector fusion reduces reliance on manual threshold-based quality control and flags sensor degradation before it corrupts records.

E ENVIRONMENTAL
Cleaner data for climate science

Higher-quality, anomaly-flagged AWS records improve the temperature, pressure & humidity data feeding climate and agricultural models.

O OPERATIONAL
Scalable, explainable monitoring

Station-level visualization, sensor-health signals and SHAP/LIME explanations support scaling to India's national AWS network.

Research & References

Core Methods & References

- Chandola, V., Banerjee, A., Kumar, V. (2009). Anomaly Detection: A Survey. ACM Computing Surveys. — basis for the statistical detector.
- Malhotra, P. et al. (2016). LSTM-based Encoder–Decoder for Multi-Sensor Anomaly Detection. ICML Anomaly Detection Workshop. — basis for the LSTM Autoencoder detector.
- Shepard, D. (1968). A Two-Dimensional Interpolation Function for Irregularly-Spaced Data. — basis for the spatial IDW consistency check.
- Lundberg & Lee (2017) SHAP; Ribeiro et al. (2016) LIME — basis for explainable anomaly contributors.
- India Meteorological Department — Automatic Weather Station network documentation.

Deliverables & Live Deployment

Codebase Reference

https://github.com/Root3141/SIH_2026/blob/main/docs/CODEBASE_REFERENCE.md

Developer Guide

https://github.com/Root3141/SIH_2026/blob/main/docs/DEVELOPER_GUIDE.md

Deployment

https://skyguardai-sih.streamlit.app/